API Documentation

Project: Airport Delay Analytics API
This document describes the RESTful API developed for managing flight data and analysing delay patterns.
The API provides the following features:

- Flight CRUD operations
- Airport destination queries
  Enter the IATA three-letter code of the airport, for example, Chicago O'Hare Airport is "ORD". "ATL" at Atlanta International Airport, etc., can show you the currently operating domestic airports in the United States.
- Delay rate analysis
  Just by entering the three-letter code of the destination airport, you can obtain the delay rate.
- Delay analysis by departure hour
  Just by entering the three-letter code of the destination airport, you can obtain the delay rate for each hour.

Authentication

This API uses a simple API key authentication mechanism.
To access protected endpoints, the following header must be included:

X-API-Key: 1122332211

Error Handling

The API uses standard HTTP status codes:

- 200 OK – request successful
- 401 Unauthorized – invalid or missing API key
- 404 Not Found – resource not found
- 500 Internal Server Error – server error

Example Request

GET /analytics/delay-rate?origin=ATL

Example Response

```
{
  "origin": "ATL",
  "total_flights": 300,
  "delayed_flights": 120,
  "delay_rate": 0.4
}
```

Airport Delay API 0.1.0 OAS 3.1

/openapi.json

default

POST /flights Create Flight

Parameters

Try it out

Name	Description
flight_number * required string (query)	flight_number
origin * required string (query)	origin
destination * required string (query)	destination
scheduled_departure * required string($date-time) (query)	scheduled_departure
actual_departure string | (string | null)($date-time) (query)	actual_departure
status string (query)	*Default value* : on_time on_time

Responses

Code	Description	Links
200	Successful Response Media type application/json Controls `Accept` header. Example Value | Schema `"string"`	*No links*
422	Validation Error Media type application/json Example Value | Schema `{ "detail": [ { "loc": [ "string", 0 ], "msg": "string", "type": "string", "input": "string", "ctx": {} } ] }`	*No links*

GET /flights/{flight_id} Get Flight

Parameters Try it out

Name	Description
flight_id * required integer (path)	flight_id

Responses

Code	Description	Links
200	Successful Response Media type application/json Controls `Accept` header. Example Value | Schema `"string"`	*No links*
422	Validation Error Media type application/json Example Value | Schema `{ "detail": [ { "loc": [ "string", 0 ], "msg": "string", "type": "string", "input": "string", "ctx": {} } ] }`	*No links*

PUT /flights/{flight_id} Update Flight

Parameters Try it out

Name	Description
flight_id * required integer (path)	flight_id
flight_number * required string (query)	flight_number

Name	Description
origin * required string (query)	origin
destination * required string (query)	destination
scheduled_departure * required string($date-time) (query)	scheduled_departure
actual_departure string | (string | null)($date-time) (query)	actual_departure
status string (query)	*Default value* : on_time on_time

Responses

Code	Description	Links
200	Successful Response Media type application/json Controls `Accept` header. Example Value | Schema `"string"`	*No links*
422	Validation Error Media type application/json Example Value | Schema	*No links*

Code	Description	Links

```
{
  "detail": [
    {
      "loc": [
        "string",
        0
      ],
      "msg": "string",
      "type": "string",
      "input": "string",
      "ctx": {}
    }
  ]
}
```

DELETE /flights/{flight_id} Delete Flight

Parameters

Try it out

Name	Description
flight_id * required integer (path)	flight_id

Responses

Code	Description	Links
200	Successful Response Media type application/json Controls `Accept` header. Example Value | Schema `"string"`	*No links*
422	Validation Error Media type application/json	*No links*

Code	Description	Links

Example Value | Schema


```
{
  "detail": [
    {
      "loc": [
        "string",
        0
      ],
      "msg": "string",
      "type": "string",
      "input": "string",
      "ctx": {}
    }
  ]
}
```


GET /airports/{iata}/destinations Get Destinations

Parameters

Try it out

Name	Description
iata * required string (path)	iata

Responses

Code	Description	Links
200	Successful Response Media type application/json Controls Accept header. Example Value | Schema "string"	No links
422	Validation Error	No links

Code	Description	Links

Media type

application/json

Example Value | Schema


```
{
  "detail": [
    {
      "loc": [
        "string",
        0
      ],
      "msg": "string",
      "type": "string",
      "input": "string",
      "ctx": {}
    }
  ]
}
```


GET /analytics/delay-rate Get Delay Rate

Parameters

Try it out

Name	Description
origin string (query)	origin
x-api-key string (header)	x-api-key

Responses

Code	Description	Links
200	Successful Response Media type application/json Controls `Accept` header.	*No links*

Code	Description	Links

Example Value | Schema

```
"string"
```

422 Validation Error

No links

Media type

application/json

Example Value | Schema

```
{
  "detail": [
    {
      "loc": [
        "string",
        0
      ],
      "msg": "string",
      "type": "string",
      "input": "string",
      "ctx": {}
    }
  ]
}
```

GET /analytics/delay-by-hour Get Delay By Hour

Parameters

Try it out

Name	Description
origin string (query)	origin
x-api-key string (header)	x-api-key

Responses

Code	Description	Links
200	Successful Response Media type application/json Controls `Accept` header. Example Value | Schema `"string"`	*No links*
422	Validation Error Media type application/json Example Value | Schema `{ "detail": [ { "loc": [ "string", 0 ], "msg": "string", "type": "string", "input": "string", "ctx": {} } ] }`	*No links*

GET /live/flights/{flight_number} Get Live Flight

Schemas

HTTPValidationError Expand all object

ValidationError Expand all object